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Course: CSA0421 - Operating System **GitHub Repo:** krishshiva721/CSA04--Assignment1

TASK 2: OS TYPE JUSTIFICATION

Architectural Selection and Architectural Trade-off Analysis

1. OPERATING SYSTEM TYPE SELECTION MATRIX

Environment	Selected OS Type	Rejected Alternatives	Technical Justification
Patient Monitors	**Hard Real-Time OS (RTOS)**	Desktop OS, Soft RTOS, Distributed OS	Requires hard deadline guarantees (< 10 ms). Standard schedulers introduce jitter; a Hard RTOS guarantees preemptive priority execution.
Nurses' Tablets	**Handheld / Mobile OS**	Desktop OS, Real-Time OS	Features aggressive power management (tickless kernel, suspend-to-RAM) optimized for battery conservation and touch UI.
Workstations	**General-Purpose Desktop OS**	Handheld OS, Real-Time OS	Supports broad peripheral drivers, complex window management, and rich productivity applications for clinical staff.
Database Server	**Clustered OS (Active-Passive)**	Single Multiprocessor OS, Desktop OS	Provides automated failover. If Node A experiences kernel panic, Node B assumes virtual IP in seconds, preventing database outage.
Imaging Server	**Symmetric Multiprocessing (SMP)**	Asymmetric Multiprocessing (AMP)	Allows all processing cores to access shared RAM dynamically, balancing heavy 3D MRI reconstruction parallel workloads.
Partner Link	**Distributed OS Gateway**	Standalone Mainframe, Local OS	Enables transparent resource sharing, identity federation, and secure cross-institutional record queries over encrypted tunnels.

2. MULTIPROCESSING ARCHITECTURE: SMP VS. AMP FOR IMAGING

Asymmetric Multiprocessing (AMP): One master CPU allocates tasks to dedicated slave processors. This creates a bottleneck on the master CPU when processing parallel 500 MB 3D MRI scans.

Symmetric Multiprocessing (SMP): All CPUs execute operating system code and worker threads concurrently out of a unified memory space. SMP provides maximum dynamic load balancing across 64+ cores, drastically reducing image rendering latency.

3. ALTERNATIVE CONCEPT & COMPLEXITY ANALYSIS

> **Alternative Scenario: Single Large Multiprocessor Machine vs. Clustered High-Availability System for Database**
>
> Replacing a multi-node cluster with a single large SMP server introduces a critical Single Point of Failure (SPOF).
>
> - **Availability Evaluation:** Single SMP machine availability is bounded at 99.9% (MTBF downtime during hardware maintenance). Active-Passive Cluster availability reaches 99.999% with failover recovery time $T_{failover} < 3\ s$.
> - **Cost & Scaling Complexity:** Vertical scaling (SMP) exhibits non-linear cost curves $O(C^2)$ due to expensive bus interconnects. Horizontal clustering exhibits linear cost curves $O(N)$.
> - **Time/Space Throughput:** Cluster memory scales linearly $Space = N \times M_{node}$, whereas SMP hits memory bus contention bottlenecks as core counts grow ($Time = O(N \log N)$ locking overhead).